

A Universe That Expands Because It Exists

A Plain Language Explainer of This Cosmology

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Why does the universe expand? Because it exists; and existing as it does, it couldn't do otherwise. That is the answer this collection of papers delivers, and what follows is the plain-language story of how it gets there. But an answer like that only means something against the one it replaces. In the standard picture the universe begins with an infinite rate of expansion and an infinite braking against it, and it survives only because the first infinity was large enough to outlast the second. The cosmological constant, the one ingredient that might have explained anything, contributes nothing at all until billions of years later. Asked why the universe expands, that picture is really answering a harder question badly: why it should exist at all.

In the cosmology I'll describe here the curve is the same — the same expansion law our telescopes measure — but nothing is supplied to keep it going. There is no initial push and no braking against it; there is one constant, present from the first instant, and the shape of a single fixed geometry read on a cosmic clock. Take that as the starting point, and the interesting question stops being why the universe expands. It becomes what the rest of it looks like: where it came from, what crossed its beginning, why it started speeding up when it did, and why the same radius that marks that moment also decides whether a group of galaxies holds together.

The horizon that never forms

Start with something familiar: a star collapsing into a black hole. The usual story has the star fall through its own event horizon, after which the horizon sits there as a permanent surface, a one-way door hanging in space.

But ask *when* the horizon forms, by the clock of anyone outside, and the answer is: never. Not at any moment of the outside universe's history, however late. A clock falling with the star reaches the horizon in a finite time of its own; the outside universe keeps every moment of its own history on

its side of that event, and the event of the horizon itself lies beyond all of them, at the outside universe's end.

This is not a trick of light, of signals arriving late. It's a fact about how the geometry orders events. Every way of slicing the outside world into "nows" piles up against the horizon without ever reaching it. And it isn't a matter of which clock you pick, either: even the descriptions built to carry a traveller smoothly across assign the horizon's events no separation at all — distinct events, in a definite order, at one place and one instant. The horizon isn't a place that exists for some amount of time. It is a single occurrence, and it happens at the end of the outside world's time — the collapsing star drawing marginally closer to it at every moment leading up to that final occurrence, in any valid description.

I proved this after a couple of years of careful thinking and writing about the relationship between spacetime and what it can mean for something to physically exist, in a sense that holds together relativistically — and about what that means for gravitational collapse in particular. Through that work I realised that a theorem could be proved in standard general relativity, with nothing added to it, that overturns the idea of a black hole that physics has carried since the 1960s: an idea that reads into the mathematics something the mathematics never demanded. That theorem is the main result of [the first paper](#) in the work this explainer describes, and it reaches further than it might seem. Roger Penrose's Nobel Prize-winning singularity theorem, which guarantees a singularity inside every black hole, starts from a surface sealed inside a completed horizon. The proof is correct. But the event horizon of every collapsing object occurs, from every outside perspective, only as a singularity of the geometry placed at the end of infinite time — so the surface the proof starts from never physically arises, and the theorem has nothing in the real universe to apply to. This past spring I wrote up [a final account of why this must be](#), and from there the rest of the picture tumbled out as I worked it through with Claude, drawing on what I'd done over those couple of years and on [my doctoral thesis](#), completed in 2012: thinking through the mathematics, and how its pieces would all fit together into one coherent whole.

So the black hole doesn't sit there swallowing things forever, as a persisting object. What sits there is matter still collapsing toward a horizon it never completes. And what lies on the other side of that completion is not "inside the black hole" in any sense a map of our universe could show. It is

something with its own time.

Where an ending turns into a beginning

Here is the central move, and it's simpler than it sounds.

At a horizon, the geometry's light-like directions take over the role that the "at rest" direction plays everywhere else. Light rays running along the horizon get stuck there; they never get anywhere. On the other side, the roles swap: what was the frozen family of light rays becomes the family of paths along which a new universe's matter advances through its own cosmic time. What the outside universe reaches only at the end of all its time is where the new universe's own time is seeded — though, as we'll see, the universe that grows from it begins further along the same curve.

Nothing has to be added for this to work. No new field, no new force, no modification to Einstein's equations. It is the same geometry, read with the roles of "light-like" and "at rest" exchanged.

And what comes out of that reading is not a universe of arbitrary shape. The geometry singles out exactly one member of its family: the one in which the black hole's horizon and the cosmological horizon of the surrounding space meet and merge — a special configuration physicists have known for seventy-five years as the Nariai geometry, and long treated as a curiosity. Read from the inside, in the frame of the matter moving along it, that geometry is a universe that is the same everywhere and looks the same in every direction, whose size grows according to one precise law. That law is exactly the one cosmologists fit to our own universe's expansion — the same curve, to the last decimal — with the timescale of the whole curve, from the first instant, set by the cosmological constant alone. What we measure when we measure how much matter our universe holds is where along that curve we happen to be standing now.

That is the content of the claim that this universe *must* expand, and at the rate we see. The expansion isn't a leftover momentum fighting a brake. It is the shape the existing universe has, given the one constant it lives in.

Flat and closed at once

Now a puzzle that has bothered cosmology for a century, and that this picture turns inside out.

Measure distances across our universe, and space comes out flat: triangles add to 180 degrees on the largest scales we can test. But a universe that came from a finite collapse ought to be closed — finite, with a three-dimensional sphere for its space, the way a balloon's skin is a finite two-dimensional sphere. Which is it?

Both. And the reason why is the most useful idea in the whole picture.

Picture the universe as a sphere of space — a three-sphere, finite and without edge. In that picture, all the universe's matter is moving: falling uniformly along a beautiful family of light-like paths that wind around the sphere in a pattern mathematicians call the [Hopf fibration](#). Every piece of matter moves at the same speed and in the same twisting sense, so no piece of matter can tell it is moving relative to any other.

Now redescribe the same situation with the matter called "at rest." In that description, space itself carries the uniform twist, and the matter sits still within it. Make that swap, and the same universe that was a sphere is described by slices that are perfectly flat — its distances those of flat space, exactly. That is what our telescopes measure.

Nothing physical has changed between the two descriptions. The same matter does the same things. What changed is which family of paths we agreed to call standing still. Flat versus curved, in this universe, is nothing but that choice. And the curvature you'd expect of the closed sphere doesn't vanish when you switch to the flat description; it reappears as the cosmological constant. Spatial curvature and the mysterious "dark energy" are one constant, read two ways.

This also dissolves the oldest question anyone asks about an expanding universe: *what is it expanding into?* Nothing. The sphere isn't sitting in some larger space with room around it. It is the whole of what exists at a moment, and its growing is the growing of existence itself. The flat description and the closed description are two honest maps of that one existing thing.

Between the parent's collapse and our universe's expansion, the two maps have to be carried through a stretch neither of them shows whole: the lap, from one seam to the other. Most of the lap runs on the cosmic clock like everything else. But one segment of it does something stranger — the

universe's size keeps changing while its cosmic time does not advance at all — and that is where the strangest and most interesting physics of the whole story takes place.

The lap: where a collapse stops and a universe begins

Earlier I said that, by the outside universe's reckoning, the horizon never forms. That is still true, and nothing here takes it back. But the outside's clock is not the only one the geometry keeps. The new universe's layer has a cosmic clock of its own, and on that clock nothing piles up against the end of time. The collapse runs its whole course at definite readings, through every stage a black hole is supposed to have — a horizon, an inside, a centre — each in a form the outside's reckoning never shows. That clock has a story to tell: the passage, seam to seam, from the parent's collapse to the beginning of ours.

Picture a single strand of that passage, and trace what its size does. Since it is always the same family of slices, we can track the whole journey by one number: the radius of the sphere that strand belongs to.

It comes in from far away, shrinking — out of the parent universe at large, in its late and thinning age, which on this clock stretches back without limit. Out there the radius keeps the time. At about twenty billion light-years it passes the first of two special radii — I'll call them seams. The first seam is the horizon the cosmological constant draws around every mass: the de Sitter horizon, the edge of the mass's own neighbourhood, past which the universe at large recedes out of reach. Crossing it inward, the radius changes character: inside, it marks a place instead, as it does in the still surroundings of a star. From there it keeps falling inward for nearly fifteen billion years of cosmic time, and the matter caught in it compresses and heats as it goes.

Then, at a radius of about twelve billion light-years, the collapse stops. On the cosmic clock the shrinking simply halts. This is the turnaround, and it is this clock's own horizon. Falling bodies come in a family, sorted by the energy they carry. The member with no energy to spare has its turning points exactly at the horizons. The member the cosmology is made of has its turning point

exactly here. So the turnaround is to the cosmic clock what the event horizon is to the outside's: the same kind of point, read by a different member of one family. On this clock there is no event horizon to cross at all, completed or not; the turnaround is what stands in its place.

What happens next — call it the lift — is the strangest stretch of the whole story, and the most beautiful. Inside a black hole's horizon, the outside's time stops being a time. Inside the turnaround, the same thing happens to cosmic time. The stretch ahead is one the falling matter has no real way across, and cosmic time turns imaginary to cross it. The radius keeps going — down past twelve billion light-years, through everything smaller, all the way to zero — and not a second passes on the real clock. What runs instead is imaginary time: the path stretches across about eighteen billion years of it. Even the shape of the descent matches the black hole's: across the lift the radius follows a sine, the kind of curve it follows inside a horizon, where everywhere else on the journey it follows a hyperbola, the kind it follows outside one.

That sounds like a magic trick, so it's worth saying what it is and isn't. Imaginary time is not science fiction. It is a tool physicists have used for decades: it describes how particles tunnel through barriers they lack the energy to climb, and it is how the temperature of a black hole was first computed. Here it does the first of those jobs, and it is not a device bolted on from outside: the lift is a barrier the collapse has no real way over, and it crosses the way a tunnelling particle does. It is the stretch of the path where the geometry's direction of time turns.

There's a neat way to see that it belongs. Along this part of the journey the motion has three characters, all from the one geometry: real at the seams, momentarily zero at the turnaround, and imaginary across this last stretch. They aren't three different laws of motion. They are one family of paths, read at three of its natural turning points.

Two cautions keep this honest. First, the eighteen billion years is a length of *path*, not of *history*. Nothing ages through it, and it doesn't make anything repeat. Second, it doesn't contradict what was said about the horizon. The two reckonings are two different clocks. On the outside's, the collapse never completes, and the horizon happens only at the end of all its time. On the cosmic clock of the new universe's own layer, the collapse reaches its own horizon — the turnaround — at a definite reading and goes on through it, and the lap is told on that clock.

At the bottom of the lift the radius reaches zero, and there the universe begins. This is the black hole's centre and the new universe's beginning at once: the place where the collapsing hole and the expanding universe meet. Every worldline turns there from collapsing to expanding — a branch point, not a wall. The curve passes straight through it. And the radius changes character once more: from here on it keeps the time, as it does inside a black hole's horizon. The lift carried the first mark of a black hole's inside, a time that stops being a time; here the radius takes up the second, and becomes the clock. That is why, in our universe, size is the clock.

Something else turns over there too. In the geometry's own terms, the radius on the collapsing side is negative and on the expanding side positive. The reflection that flips that sign, flipping the sign of the mass along with it, is the very operation that exchanges matter with antimatter at the level this geometry carries them. So the collapse our universe came from sits on the antimatter side of the branch point. In that precise sense, our universe was born from an antimatter black hole — and its own inhabitants, looking back across the same point at us, would say exactly the same of us.

Neither side is favoured. The two branches carry equal and opposite amounts of action, which sum to exactly zero. The geometry tips no scale between matter and antimatter: whatever imbalance of matter over antimatter our universe has, it brought with it.

From zero the radius grows. This is our universe's expansion, on the same cosmic clock, starting from the branch point. It grows fast at first, then ever more slowly, for about seven and a half billion years, until at about ten billion light-years it reaches the second seam, where it is growing slowest of all. Nothing about the radius's character changes here, as it did at the first seam and again at the branch point: it only touches the horizon's value and carries on. What turns instead is its pace: it stops slowing and starts speeding up, and has been accelerating ever since, for about the last six billion years.

Two numbers frame all of this, and they are not a coincidence. From the first seam to the turnaround is about fifteen billion years of cosmic time; from the branch point to the second seam, about seven and a half: two to one — and not roughly, but exactly. Draw the whole lap as one closed circuit of the underlying geometry, and the branch point divides it into a third and two thirds, 120 degrees and 240 degrees: the same two to one.

Now measure the whole journey in the time light keeps — conformal time, physicists call it — and something remarkable happens: all of it fits. The collapse, all the way in from the parent's endless late age, takes only a finite stretch of light's time; so does our expansion, all the way out into its endless future. Lay the collapse end to end with the lift, which runs at right angles to it in imaginary time, and the two make the short sides of a right triangle whose long side is exactly as long as the whole of our expansion. The sides stand in the proportions $1 : \sqrt{3} : 2$: the 30-60-90 triangle every geometry student meets. A history without end in either direction on the cosmic clock closes, on light's clock, into one clean geometric figure. And the two to one turns up again, the other way round: on light's clock the whole of our expansion is exactly twice the whole of the collapse.

Most of the physics we can actually see happened on one short stretch of all this: the seven and a half billion years from the branch point out to the second seam. Over that third of the circuit the young universe was hot, glowing and ringing with sound. It forged its first nuclei and released the light we still see as the microwave background. The rest of the lap — the parent's long collapse and the lift through imaginary time — is invisible to us directly. But it isn't silent. It decides what reaches the branch point and what doesn't, and that is where the story goes next.

What crosses the beginning

The universe's beginning is a passage, not an explosion: something goes in on one side and something comes out on the other. So the natural question is what makes it through. The answer turns out to be both very little and exactly enough.

Size crosses. Direction does not. Think of every possible pattern the collapsing universe could carry as a sum of simpler pieces: the overall size; a lopsidedness, one side bigger than the other; a squeeze along one axis; finer and finer ripples across the sphere — the way physicists break a chord into its pure tones. Now send each piece through the lift. In real time a ripple oscillates, rising and falling. In imaginary time an oscillation becomes an exponential decay instead. So across the lift every pattern that wiggles is suppressed, and the finer it wiggles the harder it is crushed.

The one piece that doesn't wiggle at all — the overall size — passes through untouched, exactly. The simplest lopsidedness comes out at under a tenth of its strength; the next pattern at a few thousandths; and from there it falls away as a plain exponential in how finely the pattern wiggles, with the rate fixed by the light's own length and nothing else — so by the time the patterns are as fine as the ripples we see in the microwave background the suppression is of order a hundred powers of ten, and it goes on steepening without limit from there. None of these numbers was chosen; there is nothing in the geometry to adjust. And a second, independent route gives the same answer: the radius where the parent's horizons merged is a pure de Sitter region, and de Sitter space is known to forget everything but its own scale.

That is why the sky is smooth. One of the great puzzles of standard cosmology is that the microwave background looks the same, to one part in a hundred thousand, in every direction — including regions that, on the standard account, could never have exchanged a signal. The standard fix is inflation: a separate, adjustable burst of expansion added to stretch away the differences. Here nothing is added. The beginning itself is an isotropizer. Whatever lopsidedness the parent's collapse carried in, the passage hands on only its size. A smooth sky isn't a coincidence to be explained. It is what a universe that began this way has to look like.

The slate is melted clean. As the parent's matter falls inward on the collapse leg, it heats — eventually past the point where atomic nuclei can survive. Whatever the parent had built, its heavy elements, its stars, its chemistry, is broken back down into free protons and neutrons. Then our side of the branch point cools back down through the same temperatures, and ordinary nuclear physics forges the first light nuclei again, from scratch. Run that, and helium and deuterium come out at the amounts astronomers measure. The oldest stars we see are almost free of heavy elements, which is exactly what a slate wiped clean requires. Lithium comes out at the same threefold excess the standard model predicts — that puzzle sits in the nuclear physics both pictures share, not in this geometry.

The sound starts together. The young universe rang like a bell: pressure waves sloshing through the hot plasma, which we now see frozen into the microwave background as a series of peaks. Those peaks are sharp, and that is a clue. Sharp peaks mean every sound wave of a given wavelength started at the same moment, in step; if they had started at scattered times, the peaks

would have washed out into a smear. In the standard picture that coordination is another job handed to inflation. Here it comes from the beginning itself. The sound waves of the parent's collapse don't make it through — the observed peak positions independently require that they don't — and every sound wave on our side starts fresh, together, from the branch point. One beginning, one shared starting gun.

And the parent hands over a small inheritance. Some things the geometry does not make, and the useful question is where they come from instead. The proportion of matter to light; the slight excess of matter over antimatter, since the passage itself favours neither; and the overall strength of the faint ripples that later grow into galaxies, together with their slight tilt toward larger scales. These come from the parent, handed across at the beginning — the way a child inherits a few particulars from a parent while the shape of a human body comes from something deeper. Standard cosmology treats the first two the same way, as measured inputs, so nothing is lost by it. And that the tilt belongs on this list is settled by counting rather than by elimination. Trying candidates one at a time — something in the geometry that might produce a tilt after all — can never finish, because there is always another candidate to think of. Written out properly, the pattern that emerges is a product of just three things, and a tilt can only come from a term that varies with scale. One of the three is a plain constant and cannot tilt anything. That leaves exactly two, and both are already accounted for: what the collapsing interior multiplies the pattern by, which is itself a constant here, and what the parent handed over. So the tilt is the parent's not because the alternatives have run out but because there were only ever two places it could come from.

The ripples turn out to be more interesting than the standard story makes them. On the inflationary account they begin as quantum jitter in empty space, stretched to astronomical size. Here that can't be their source: the jitter this geometry's empty space supports is weaker than the observed ripples by more than a hundred orders of magnitude. The seeds of the galaxies are not quantum noise blown up. They are real, classical structure, the parent's own. And the lift, for all that it erases direction, turns out to dim them rather than wipe them out: what crosses is the parent's pattern, faint but not gone, with the broadest lopsided shapes dimmed least of all. How strong the pattern was to begin with, and its slight tilt, the geometry does not say. That is the parent's to hand over, and the work takes it as given, just as the standard account does — the one part of the ripples' story still carried as an input rather than derived.

There is one more thing on that route. Of the two seams, the first — the de Sitter horizon around the mass — is the one that behaves like a horizon in the full sense: it has a real strength to it, where the second merely touches the horizon's value and carries on. And the ripples get their pattern well before they reach it: far out in the parent's thinning age, in the first per cent of the fall. Then they cross it. A horizon with a real strength has a temperature, and a temperature can stamp a pattern of its own on whatever lingers near it — which is exactly how the inflationary account makes its ripples. So the question was whether this one restamps the parent's.

It does not, and the reason is clean. At a horizon the geometry picks out two families of paths: ones that cross straight through, and ones that run along it and never quite arrive. The temperature belongs to the second family — it is what the endless not-quite-arriving amounts to. The ripples are on the first: they come in at speed and go through. The two never trade. Work the wave equation out at the seam and the factor the crossing applies to a wave is exactly one — not nearly one, not one for long waves, but the number one, for every wavelength. And the scale at which the horizon's temperature could have acted at all, had anything been listening, is larger than the largest wave the sky contains: below even the broadest lopsided shape, a factor of fifty beyond the first sound peak.

That factor of fifty keeps turning up, by three routes that have nothing to do with each other. Fifty in radius — the ripples take their pattern around fifty times further out than the horizon they later cross. Fifty along the path, measured in the time light keeps. And fifty in wavelength, between the scale the horizon's temperature sets and the first sound peak. Three different quantities, agreeing to within a tenth. When a number arrives three ways like that, it has stopped being an estimate and become a feature of the geometry.

There is one more way the horizon's temperature might matter. A crossing is one thing; a horizon that simply *glows*, pouring its own radiation into the universe forming behind it, is another, and that would not care whether anything was crossing. But to work out what a horizon radiates you need its temperature to be a single definite number, and here it is not. The two seams are one place in the underlying geometry, met a lap apart — and that one place carries two different strengths, a real one at the first seam and zero at the second. A temperature has to be *the* temperature of *a* horizon; this one has two, and nothing reconciles them. The second reason is independent and just as plain:

the calculation would have to compare the lap's own clock with the clock the temperature is defined in, and the conversion between those two clocks falls to zero at exactly the two places the comparison would have to be made.

So this is not "the horizon glows and the glow turns out to be harmless." It is that the construction never hands you the object a glow would be a property of. Worth keeping the distinction: the first would be a mechanism, something to count; the second is the absence of the thing a mechanism would be about.

So the parent's pattern arrives as the parent sent it. The chain from the old universe to our sky is closed end to end, and the one thing still carried as an input is the strength and tilt the parent handed over in the first place.

Why it speeds up, and why then

In 1998 two teams of astronomers, measuring distant exploding stars, found something no one had expected. For roughly the first half of its history the universe's expansion had been slowing down, as you'd expect if gravity were pulling everything back together. Then, about six billion years ago, it stopped slowing and started to speed up, and it has been accelerating ever since. The discovery won the 2011 Nobel Prize.

The standard explanation adds a new ingredient, dark energy: a kind of energy belonging to empty space itself, whose density stays fixed while matter thins out as the universe grows. Early on matter dominates and its gravity brakes the expansion; once matter thins to half dark energy's density, the balance tips and the expansion accelerates. That account works. But it treats the turn as the outcome of a tug-of-war between two unrelated things, and it leaves the moment of the turn as an accident of how much of each happened to be there.

In this cosmology there is no tug-of-war, because nothing is pulling and nothing is pushing.

A bend in a fixed curve. The expansion here is not a running tally of impulse against braking. It is the shape of one fixed curve — the universe's size, read on its own cosmic clock. A curve can bend one way and then the other. Think of a road that curves downhill-concave and then uphill-concave,

with one point in between where it runs momentarily straight. Mathematicians call that point an inflection. The universe's expansion has exactly one. Before it, the curve bends so that growth slows; after it, so that growth quickens. At the inflection itself the universe is growing more slowly than at any other moment in its history.

And the bend is not a metaphor. Each moment's slice of the universe is a surface cut through the underlying geometry, and that surface has a curvature of its own. The universe's acceleration at each moment turns out to be exactly that curvature, multiplied by its size. The mass bends the slice one way; the cosmological constant bends it the other; where the two cancel, the slice is momentarily flat — and that is the inflection. The slowing and the speeding up are not two forces taking turns. They are the two sides of one bend. Nothing accelerates anything; the change from slowing to speeding up is a property of a fixed curve. And the standard account's own condition, matter at twice dark energy's density, falls at exactly the same moment. The numbers agree; what differs is what they mean.

When, and at what size. That inflection came about seven and a half billion years after the beginning, some 6.3 billion years ago — light from the galaxies we see at redshift 0.68 left them at about that moment. But here this cosmology can say something the standard one cannot even ask. In the standard picture the universe's "size" is a bookkeeping number with no absolute value; only ratios of it mean anything. Here the size is real — the radius of an actual sphere — and so the turn happened at a definite one: about 9.8 billion light-years, which is exactly one over the square root of the cosmological constant. The universe turned from slowing to speeding up when it grew past that radius.

And that radius is not just any number. It turns out to be several things at once.

First: it is the universe's own holding radius. Take any mass in this universe — a galaxy, a group of galaxies, a cluster. Its gravity pulls things toward it; the cosmological constant carries things away with the general flow. There is a radius around any mass where the two balance: inside it, the mass can hold things to itself; outside it, the flow wins and carries them off. It is called the Hubble–Eddington radius. For a galaxy like the Milky Way it is about three and a half million light-years;

for our Local Group of galaxies, a little under five million; for the great Virgo cluster, about thirty-five million. Astronomers can see this boundary around nearby groups, where galaxies stop falling inward and start receding with the flow.

Now take the mass of the universe itself and ask the same question. The answer is 9.8 billion light-years — the radius at which the expansion turned. The universe began to accelerate at the moment it grew past the size its own mass could hold. The rule that decides whether a galaxy group holds together is the same rule that decided when the whole universe let go. The universe is the largest structure there is, and it obeys the same boundary as every smaller one — it just happens to be the one structure that lives inside its own.

Second: it is the radius of the parent's merged horizon. Go back to the special configuration the parent's collapse selects — the one in which the black hole's horizon and the surrounding cosmological horizon meet and merge. They merge at a radius, and it is this one. It is the one member of the whole family of black holes in which the holding radius lies exactly on a horizon; and in that same configuration, it is also the radius at which light can circle the black hole in a closed orbit. Three distinct properties of the parent — where its horizons merge, where light can orbit, where its gravity and the cosmological constant balance — all sit at one radius. And our universe began to accelerate as it passed through it.

Third: it is where the universe takes the horizon's shape. The universe's space is a three-sphere, carrying the uniform Hopf twist described earlier. Follow that sphere toward this radius, described the way that keeps the sphere in view, and it doesn't stay round. It flattens along the direction of its twist, more and more, until in the limit the twist direction closes up entirely, and what remains is a two-dimensional sphere of radius 9.8 billion light-years — the very sphere of the parent's merged horizon. What becomes of the remaining direction is just as exact: it becomes the geometry of the thin region between the two merged horizons, the configuration physicists call the Nariai throat. The throat and the universe's own space, at that radius, are one object described two ways. On its way from slowing down to speeding up, our universe passes through the precise shape of the horizon it came out of.

So the answer to "why does the universe speed up, and why then?" is this. It speeds up because the curve it lives on bends that way past a certain point, and nothing more is needed. It does so then because that point is where it outgrows the hold of its own mass — the same radius where its parent's horizons merged, and the same shape. The one constant that sets how fast the universe grows also sets the size at which a galaxy group keeps or loses its outermost members. In this picture they are not two jobs. They are one rule, read at two scales.

How far down does it reach?

So far this has been a story about the largest things there are: universes, horizons, the shape of space. It's natural to ask whether the same geometry has anything to say about the smallest — the particles everything is made of. It does, and what it says is precise. It also stops where it stops, and where it stops is as interesting as where it reaches.

What the geometry doesn't do. The great hope of a century of physics has been to find the forces of nature written into the geometry of space and time — gravity is geometry, so perhaps the strong and weak nuclear forces are too. The natural place to look in this cosmology is in the motions the underlying geometry allows, its symmetries. That door is shut, cleanly: the symmetry that holds quarks together, called colour, is simply not among the motions this geometry can make.

But the geometry has a second face. Read with time turned imaginary — the same move that appeared on the lift — the same mathematics becomes a timeless, finite, perfectly round space, and colour lives there. So does the scale of the quantum, Planck's constant. So the split that has dogged physics for a century — general relativity on one side, the quantum and the strong force on the other — might not be two theories waiting to be glued together. It may be one geometry, seen on its two faces: one carrying time, gravity and matter; the other carrying colour and the quantum.

What the geometry does do, for matter. Matter, though, does live on the face with time, and there the geometry delivers three things outright.

Put the simplest kind of matter particle into this geometry — the kind electrons and quarks are made of — and ask where it can sit. It binds at a wall: the place where the geometry's radius changes sign, which is the very place where, read cosmologically, a universe begins. One particle state binds at each such wall, and it comes out with a definite handedness, as if it spins only one way — a famously strange feature of real matter, which nature's laws distinguish from its mirror image.

How many walls are there? The underlying geometry is symmetric in three ways, set 120 degrees apart — the same 120 degrees that splits the lap into a third and two thirds. The least arbitrary way to build matter on it uses all three, so there are three walls, and three particle states, all of the same handedness. Real matter comes in exactly three families — the electron and its two heavier cousins, the muon and the tau, and likewise for the quarks — and nobody knows why there are three. Here the number isn't put in: it is the geometry's own threefold symmetry, and no loop or twist of the geometry can change it. And there is a further surprise. The construction only works with one time and three space dimensions: four is the only number of dimensions that can carry both a count of families and a handedness. In this picture, three families of matter and four-dimensional spacetime are one fact, seen from two ends.

Go one step further and let three of these particles combine, and the geometry selects exactly the combinations nature allows for quarks — three together, or a quark with an antiquark — and forbids the ones nature forbids. It does this without supplying any force to hold them: it delivers the rules of colour but not the glue.

Where it stops. It doesn't give the particles their masses — in nature those come from the Higgs field, and here they still do. It doesn't give their charges either, and it falls short on one detail of the right-handed particles, which the ordinary gauge route the Standard Model already uses supplies. That's the boundary, and it's drawn sharply. The geometry fixes the skeleton of matter — how many families, which way they spin, how they may combine, and their antimatter partners — not its flesh. The same three-way symmetry that places the horizons and the seam also places the walls where particles bind. Whether that shared root runs deeper, into the forces themselves, is a question this work has not answered.

The far horn

Now look forward. Our universe passed its seam six billion years ago, and on its own curve it will go on accelerating forever. Its growth settles toward a steady pace: the size of the universe doubles about every twelve billion years, set, again, by the one constant alone. Distant galaxies slip one by one beyond the reach of our light, and what stays bound to us is what lies inside our own holding radius. This is the far horn of the curve: the open, ever-widening region the expansion runs out into — in the language of the geometry, the de Sitter region our universe tends toward.

But our universe is not empty, and it is not quiet. Stars burn out and collapse. Galaxies grow black holes at their hearts. And now the whole story turns over.

None of those collapses ever completes a horizon — not by our clock, and not by any other way of reckoning time. A completed horizon is something the actual universe never contains: what exists at every moment is matter still collapsing toward it. That is why the famous puzzles about black holes evaporating and destroying information never get started here: they presuppose a completed horizon that no moment of our world contains.

But collapse, on the geometry's own terms, cannot end in nothing. It continues as a cosmology. Every black hole forming in our universe is, from its own side, the beginning of a universe: the same passage, the same lift through imaginary time, the same branch point, run again.

And there is a pleasing symmetry in it. Our universe came from the antimatter side of its branch point. By the same reflection, the universes our own black holes open onto sit on the far side of the matter–antimatter mirror from us. To their inhabitants, it is we who are made of antimatter.

So the universe is not a one-off. It is one generation in a lineage: each universe born from a collapse in its parent, growing according to the same rule, and becoming a parent in turn wherever its own matter falls together. What passes down the line is very little — the size, the smoothness and the shape come from the geometry every time; what is handed on is a small inheritance of particulars. Whether the cosmological constant itself is among them — whether a child universe necessarily lives in its parent's constant — is one of the open questions, and either answer would tell us something.

What the shadows are of

Go back to the question we started with. Why does the universe expand?

In the standard picture the honest answer is: because it was given an enormous start and hasn't yet been stopped. Expansion is a holdover, an impulse from the beginning still outrunning the drag of everything since. Behind that answer sits a harder question it can't touch: why there should be a universe that lasts at all, rather than one that falls back before anything has time to happen.

The picture laid out here answers differently, and the difference is the whole point. Nothing was given a push. There is one constant, present from the first instant; a layer of existence advancing through cosmic time; and the shape of one fixed curve, which is simply how that layer's size reads on its own clock. The universe doesn't expand in spite of anything. It expands because it exists, and because existing as it does, it couldn't do otherwise.

Everything else in this story has turned out to be that same fact, seen from somewhere new. The beginning: a collapse that cannot end in nothing, and so carries on as a universe. Space that reads as flat or as a finite sphere depending only on which paths we choose to call still. A stretch of imaginary time that hands on a universe's size and wipes out its lopsidedness, which is why the sky is smooth. A clean slate of composition, and a single starting gun for every sound wave. A turn from slowing to speeding up at the size the universe's own mass could hold — the radius where its parent's horizons merged, and the same rule that decides whether a group of galaxies keeps its outermost members. Matter's three families and its handedness, seated on the same threefold geometry as the horizons, at the very wall where a universe begins. And a lineage: every collapse in our universe, on its own side, a new beginning.

And here is the part that still astonishes me. Wherever this picture meets the measurements, it keeps landing on the standard numbers — and every time, for a reason of its own. The expansion follows the same curve, with no push behind it and no brake against it. The light elements come out where they're observed, cooked not in a hot beginning but in a collapse's own heat, wiped clean at the turn, and cooled back through the same narrow window at the same rate. The ripples come out nearly alike at every scale, not because something smoothed them flat, but because the passage they cross has no scale of its own to bend them by. And the microwave background rings with the same

comb of sound peaks cosmologists have spent decades measuring — from classical structure, not quantum jitter, every wave started together at a single boundary, the spacing landing to one part in six hundred with nothing tuned to put it there. No second theory is laid over the first to make any of this happen, and no needle is pulled from a haystack of possibilities. The geometry, run forward, simply does it.

None of these pieces was added to make one of the others work. Each one turned up when the one before it was looked at closely: the same results, arriving by different roads, and the roads all turning out to be one. That is the feature I find most persuasive, more than any single result. The picture keeps paying out more than was put into it.

The test that matters, of course, is not whether a picture is pretty but whether the universe agrees with it. Here the agreement is already substantial. The expansion history this geometry requires fits the most precise maps of the universe's growth we have; fixed by those maps and the acoustic scale together, it puts today's expansion rate near 68.6 kilometres per second per megaparsec. The light elements come out where they're measured, with lithium off by the same factor of three it is in the standard account. The spacing of the microwave background's sound peaks, computed with nothing adjusted to fit them, comes out right to about one part in six hundred. What doesn't yet line up is the finer shape of those peaks — how strongly each rises against its neighbours. Give this cosmology's spectrum the same freedom the standard model is normally judged with — its expansion rate, its two densities and the tilt of the primordial ripples all fitted to the microwave data — and the standard model sits essentially on top of the measurements while this one sits at about one and a half times their distance from it. The freedom closes about a third of the gap and leaves the rest, and the rest is what the paragraphs below take apart. It is not in the sound itself: the pattern as it stands on the early sky comes out as it should. It enters in how that pattern is carried across the distance to us, and it turns on one choice — which of the geometry's two rates that distance is built on. The two differ by exactly one thing: the plasma's rate carries the radiation and the geometry's does not. Build the distance on the plasma's rate and the discrepancy disappears entirely — but that is the same as putting the radiation back into the expansion, and the light elements shut that door independently: helium comes out too high and deuterium far too low. So the one lever the sky appears to be pulling is closed by other measurements. And the obvious next thought — that some parameter is simply set wrong — is answered: give this cosmology's spectrum its head, with the

expansion rate, both densities and the tilt free, and it settles within a twentieth of a standard deviation of the background the distance measurements had already fixed, at a cost of one part in a hundred. The sky is not pulling it anywhere. What is left over is therefore not a number anyone has failed to adjust; it is a shape, under a per cent in size, that no free parameter absorbs.

The shape is specific. The leftover wiggle runs in step with the sound peaks, but when its own spacing is measured rather than assumed, it comes out about five per cent longer than theirs: 312 against 298, with 298 excluded at better than a thousand to one.

That does not mean the ruler is off. The peak spacing itself is right to better than two parts in a thousand, and the offset is more than thirty times that spacing's own uncertainty. You cannot get there by nudging a length.

What it is instead is something cleaner. A wiggle at the wrong spacing, and a wiggle at the *right* spacing whose phase slowly slides as you go to smaller scales, are not two possibilities — they are the same thing written two ways, exactly. So the residual is a slow slide of the peaks' phase: about a hundred degrees of drift across the whole range we can see, roughly a quarter of one peak-to-peak cycle.

A sixth of that hundred degrees is not the universe's at all. The same act of projecting the sound onto the sky that makes the gaps drift also slides the phase, by about fifteen degrees — and because that slide depends only on lengths the two pictures share, it is the same in both and cancels when one is set against the other. (A sixth is the figure for the stretch of sky we can actually see; over a different stretch it would be anywhere from a tenth to a quarter, so the fraction has to be quoted with the range it was read over.) Which also closes off projection as the cause of the remaining five sixths: the difference the projection makes *between* the two pictures is ten thousand times smaller than the leftover being explained. So the candidate causes stay at three, and the thing needing a cause is a little smaller than it looked.

What that slide *is*, though, the measurement that found it cannot say — whether it belongs with the small, constant offset in the peaks' phase this work had already measured, or is something else. The trouble is that a slow slide across the whole range looks the same whether the peaks' phase is really sliding or the overall spectrum is merely tilted a little. You can see the measurement cannot separate

them, because it reacts just as strongly to a knob that *provably* cannot shift a peak at all: the primordial tilt, which multiplies the initial power and leaves the sound physics untouched. So the drift's *size* is solid — the two descriptions really are the same thing written twice — but its *name* is not something that measurement can supply.

A measurement that can tell them apart looks at where each peak actually falls, after dividing out one overall tilt. Two numbers come out — how far the whole set sits shifted along, and how far the odd-numbered peaks sit out of step with the even ones — and between them they separate a real change in the sound, such as how heavily ordinary matter loads the plasma, from a mere tilt by factors of eleven and twenty-one, where the old measurement managed one and a third, which is next to nothing at all.

On those two numbers the two candidate causes part cleanly. Computing the model's spectrum with the driving switched off shows them pushing the peaks in *opposite* directions — more driving slides the whole set one way, more loading slides it the other — and differing in strength by a factor of five, which is a far stronger kind of distinction than one simply being bigger than the other.

There is a third possibility. This geometry carries two natural clocks, and they disagree about the very stretches the sound is counted across — not slightly, but by a factor of four and a half, to the point of disagreeing about which of the two stretches is the longer. So the sound waves could be being driven on the wrong one. That would leave the spacing untouched — the sound horizon is an integral taken on the waves' own clock — and could only move the phase, which is the shape the leftover wiggle actually has. On the two numbers, a wrong clock is separated from the loading by a factor of six, but not from the driving: an answer on the driving's side would mean *the driving, or the clock the driving is read on*. Where those two part company is further out. A wrong clock does almost all of its work on the sky's broad patterns and then goes flat, while the driving's own phase turns over and keeps moving right out to the finest scales measured — so weighing the sky's fine detail against its broad patterns is what would split them.

All of that compares one calculated curve with another — what each cause *would* do. The sky's own leftover wiggle can be placed on those same two numbers only after one change to how the data are read, and the reason is specific. The method finds where the peaks are and then measures exactly where each one sits. On a smooth calculated curve there are five peaks and it finds five; on

the real sky, noise makes bumps, and it finds twenty-one, settling on three that are not separate peaks at all but ripples on the shoulder of one. Resampled over the data's own uncertainties it fails outright forty-nine times in fifty. So the answer is not uncertain; the tool is not measuring the thing it is pointed at. And the reason is narrower than "the sky is noisy": the sky is reported in bins just narrow enough that the scatter within one bin is a little over twice the real change from one bin to the next, which is exactly the condition under which a bump-finder cannot tell a peak from a wobble. Hold on to that, because it is where this ends up.

The other way to go about it is to stop hunting for peaks one at a time and fit a whole regular comb of them to the data at once. On six of the model's own spectra, a comb with a fixed spacing reads three of them correctly and misses the other three by about a fifth — and which three it reads is not a matter of whether the driving is on. The reason is in the gaps between the peaks: they do not just alternate, they grow. The spacing itself drifts, and a drifting spacing is exactly a slow slide of phase — the very shape the leftover wiggle was found to have. Where the drift comes from is known, and it is not the sound: the sound's own peaks sit on an exactly even ladder, and the drift is manufactured by the act of projecting that ladder onto the sky, which smears each peak asymmetrically and nudges it outward by a shrinking amount. A perfect evenly-spaced comb with no physics behind it acquires the whole effect on the way out. That makes the drift something this cosmology predicts with nothing fitted — and also something the standard account predicts identically, since the three lengths it depends on are shared. Where that drift is mild the fixed-spacing comb copes; where it is not, the comb fails, and that is the whole of the division. So the comb needs that one further freedom before it can be pointed at the sky.

Given the freedom, it turns out to be the right one — on four of the six curves the drift the fit recovers agrees with the drift the gaps show, in sign and to within a sixth, on a comparison it was not tuned to make — and the data turn out not to be able to hold it. The reason is sharper than it first looks. It is not that a spacing and a slow change in that spacing blur together: they come apart cleanly, and all it takes is to quote the spacing from the middle of the stretch of sky being used rather than from its edge. Do that and the two separate exactly, leaving the fitted curve untouched. But what the data then pin down is the spacing of the peaks in the neighbourhood of the second one, and that is not the number this comparison is about. The number it is about lies in the other direction of the same pair, and that direction is loose. Worse, the tightly held one turns out not to be

about the sound at all: on the two curves where the comb has a real grip on the peak train it is just that train's average gap, and on the four where it has lost its grip — the real sky among them — it is a fifth to a quarter away from any gap the curve actually has. So it measures whether the comb is holding on, which is not a question about the universe.

And the blurring that caused all this is not a shortage of data. It is geometry: a level and a slow slide in the same quantity are the kind of pair whose overlap is fixed by the ratio of the two ends of the stretch you look across, not by how many points you put inside it. Halve the number of points and the overlap changes in the fourth decimal place. So more points would not help, and the honest end of this route is that it is closed for a reason rather than for want of resolution.

What does help is the opposite of more points. Group the sky's bins in pairs — the scatter within a bin then drops to about four fifths of the real change between bins, back on the right side of the condition noted earlier — and the plain peak-finder reads the first four peaks of the real sky to within three or four units of where it reads them on a calculated curve treated exactly the same way: gaps of 303, 268 and 309 against 302, 269 and 312. Group them in threes and it finds nothing at all, because its own search window then holds too few points to work with. So the useful step is a factor of two and only a factor of two, and the two instruments want opposite things: the comb wants more points and cannot use them, and the peak-finder wants fewer and can. The route to the sky is the peak positions on the coarser grouping, at four peaks rather than five — and what that route has to show is whether the two candidate causes still part cleanly on four peaks, when the separation between them was established on five.

They do. Carried onto the paired-up data, the two-number measurement reproduces the driving difference it was built on to within one per cent, and the two candidate causes still part by a factor of five and a half. The separation from a mere tilt falls from eleven and twenty-one to between eight and twelve — weaker, but still most of an order of magnitude above the one and a third the old measurement managed — and what is lost is lost in the control rather than in the signal: with one peak fewer, the tilt leaks four times as much onto the second number.

So the sky goes on the plane, and this is the first thing this line of work says about the actual sky rather than about an instrument. The sky sits about eight tenths of the measurement's own uncertainty from the matching calculated curve — on top of it — while one full unit of extra or

missing driving would sit thirteen uncertainties away. So the sky is consistent with the driving the calculation already carries, and a change of one whole unit either way is ruled out. It is the driving this bounds: per unit, the loading moves the measurement only about a seventh as far, so the sky says much less about that.

Three things limit this, none of them noise, and they cut the two halves of the statement differently: the exclusion holds, and the agreement is the provisional half. The reason is what each half compares. The thirteen compares two calculated curves, so anything they share cancels out of it. The eight tenths compares the sky with one calculated curve, and there nothing is guaranteed to cancel. First, the fifteen degrees the projection manufactures: between two calculations it cancels to a part in a thousand, but between the sky and a calculation it cancels only as far as the calculation's projection is the real one — and it is nearly six times the displacement measured, so if none of it cancelled, the sky would sit a little over five uncertainties out instead of one. Nor is that a gap waiting to be filled: settling it would mean knowing whether the calculation's projection is the sky's, which cannot be had from the same sky without arguing in a circle and from anywhere else is some other calculation's answer. Second, the peak spacing is assumed rather than measured: the agreement needs it right to four parts in a thousand, where the exclusion needs only five per cent — and that is not something this measurement can settle either way, because a spacing read off the same peaks is already inside the thing being tested, while a spacing brought in from outside is a standard-model fit to the same sky.

The last limit is of a different kind. One of the peaks may not be there to find — and it is the third peak, not the fourth: in about half of the simulated repeats of this sky the finder comes back with three instead of four, and in all but a twelfth of those the peak it misses is the third, the fourth being found in 96 of every 100. So the measurement exists on about half the ways this sky could have come out. The first two limits bear only on the eight tenths; this one could have borne on both. Every uncertainty here is the spread inside the half of the repeats that yield four peaks, and if those were the repeats that landed closest in, that spread would be too narrow and the thirteen too large. It is not: the spread inside that half is 0.98 of the spread across all of them, and the repeats thrown away are slightly wider rather than narrower. So what the missing peak costs is repeats, not width, and the thirteen stands where it is quoted.

The whole account rests on a single identification: that the distance the projection uses is the one the observer's own flat slice reads. It has been examined by looking for a second, independent way to reach the same distance, one that doesn't mention a slice at all.

There is exactly one such way: follow a bundle of light rays and watch how the geometry focuses it. Carried out, it does return the flat distance. But the rays it returns it for are the wrong rays — light on them comes in *blue*-shifted, by a factor ten thousand out and in the wrong direction, where the real sky is redshifted by a thousand. So it is not a second route to the answer; it is a different question with a coincidentally familiar answer.

And the reason no second route exists turns out to be the better finding. The distance is made of two pieces read off two different things — a length measured across the observer's flat slice, and a redshift read from the layer's own size. The flat slice is a purely spatial surface; no light travels within it as such, so there is nothing there for a light-bundle argument to be about. Which means the identification was never the kind of claim a second support could strengthen: under the reassignment, the path light takes simply is a path in that slice. The distance is read there because of what light is here, not because of an inference that might have been checked twice. A narrower position than three routes, and a firmer one — it cannot be seconded, and equally it cannot be undermined by the absence of a second. What remains is the shape itself.

There are sharper tests to come. One is the boundary around groups and clusters of galaxies — the holding radius — where this geometry says exactly how far a mass can keep hold of its neighbours against the flow. That one has been waiting on the sky rather than on the theory. Testing it needs two things measured on the same cluster: where the boundary falls, and how much mass lies inside it — and found independently of each other, which no existing catalogue manages. The samples that looked usable had fixed their outer edge by definition rather than measuring it, so the answer was built into the question before anyone asked it. A survey designed for the purpose changes that: about a hundred clusters, mapped out to five times the radius usually taken as a cluster's edge, with several thousand galaxy velocities in each — enough to weigh what is inside the boundary and to find the boundary from the motions themselves. It is the first sample that could return a real answer, and what it will meet is a prediction with nothing adjustable in it. Five times the cluster's edge is where that survey chose to stop; it is also, as it turns out, where this geometry puts the boundary —

5.23 times, from one line of algebra with no fitted quantity in it. The survey picked its reach for its own reasons and landed on the number the theory had been carrying all along. The claim at the root of it all needs no test of its own, because it isn't a hypothesis. That no black hole ever completes its horizon follows from general relativity itself, from the same structure every black-hole calculation already uses, read without assuming the conclusion. It changes nothing anyone outside could ever see; what it changes is what we take to exist. What the sky can judge is what comes after: if collapse carries on as a universe, then ours should look like one from the inside — flat, smooth, expanding by one law with one constant setting its pace. That is the universe our telescopes find.

This is a picture, finally, about what the shadows are of. For a century the four-dimensional block of spacetime has been treated as the thing itself — the whole of what exists, laid out at once. But a map of everything that happens is not the same as what exists. Spacetime is the record; what exists is the layer advancing through it. Read the record as a record, and the puzzles that came from mistaking it for the world begin to fall away: why there is a beginning, why space is flat, why the sky is smooth, why the expansion turned when it did. What's left is a universe that was never in danger of failing to exist — born from collapse, growing by a rule set by one constant, and giving rise to others like itself. It expands because it is there.